

CDI LABS

Empowering Discoveries in Disease Research

CDI Labs products and services

BROCHURE

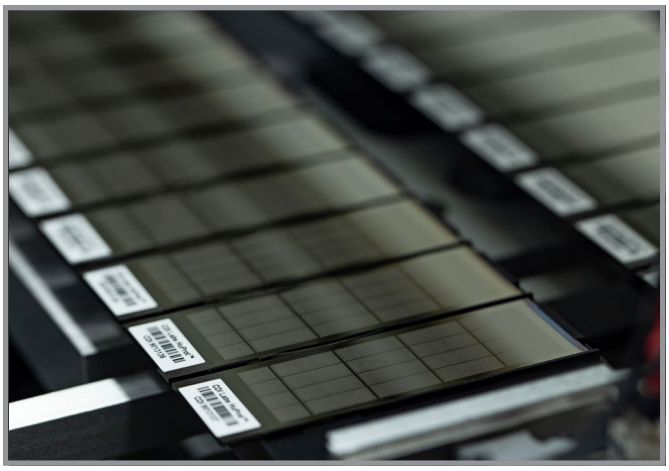
JUN 2024

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HuProt™ Human Proteome Microarray

The Most Powerful Tool Ever Created for Autoantibody Profiling and Target Identification

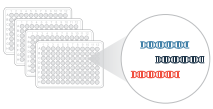


HuProt consists of 21,000+ unique human proteins and isoform variants – covering 16,794 unique genes. This includes 15,889 of the 19,613 canonical human proteins described in the Human Protein Atlas, with broad coverage across protein subclasses.

Most fluids contain antibodies and we can analyze them (*i.e.*, peripheral serum/plasma, cerebrospinal fluid, tissue lysates, stool, saliva), or in a synthesized sample, for any isotype (i.e., IgM, IgG, IgA, IgD, IgE). Secondary antibody binding is then quantified using fluorescence intensity measurements of features on the array. We'll provide you with those RAW readouts with normalized readouts and group-comparison statistics (optional). HuProt arrays also allow for pan-isotype autoantibody biomarker profiling versus properly folded, three-dimensional human proteins.

HuProt Library Curation and Production

There's more to HuProt than serum profiling. We've profiled nucleotide binding, antibody specificity, small molecule binding, protein-protein interactions, and more. If you can dream up a study, we can help you get it done. HuProt contains >21,000 unique proteins and was built during the NIH Protein Capture Reagents program.



1

Protein-encoding open reading frames (ORFs)

synthesized or obtained from public ORF libraries



2

Gateway cloning in *E. coli*

individual ORF DNA sequences were shuttled into a yeast high-copy expression vector (pEGH-A) that produces N-terminal GST-fusion proteins



3

Protein-encoding vectors rescued from *E. coli*

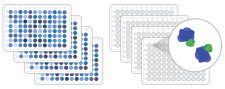
and used to transform *S. cerevisiae* yeast



4

Transformed yeast clones bi-directionally sequenced

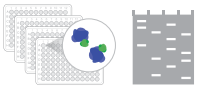
to confirm correct ORF insertion



5

Sequence-confirmed yeast clones are grown, lysed, and harvested

GST-fusion proteins are purified via GST-binding agarase beads



6

Confirm purity and expected molecular weight



7

Arrays are printed



8

Slides are stained with anti-GST and scanned

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HuProt Methods and Analysis

HuProt is an incredibly sensitive antigen matrix platform that provides quantitative binding interaction data from a few microliters of sample. It has been used to identify everything from the immunologic fingerprint of cancer patients destined for checkpoint blockade toxicity to novel protein-protein interactions.



1

HuProt is reliable

Sequence-confirmed GST-fusion protein from a library of 21,000+ full-length human protein is individually synthesized, purified, and printed

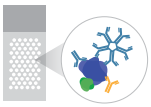


2

HuProt is versatile

Detects most target of interest via fluorescent labeling.

Any antibody type (IgM, IgG, IgG1, IgG2, IgG3, IgG4, IgA, IgA1, IgA2, IgD, IgE), custom protein, nucleotide, and more

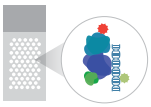


3

Serum and Plasma

Control samples are diluted 1:1000 in blocking buffer and reacted with arrays

OR



3

Protein and DNA/RNA

Fluorescent labeling for protein or sequence of interest



4

Array scanner captures raw fluorescence data

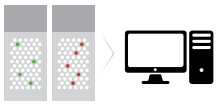
of sample antibodies of interest



4

Array scanner captures raw fluorescence data

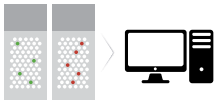
of labeled targets of interest



5

HuProt library file aligned to capture spot readings

Arrays passing spot QC proceed to case/control anaytics pipeline



5

HuProt library file aligned to capture spot readings

Arrays passing spot QC proceed to target Z-score pipeline

Technical Details

With 21,000+ human proteins and protein isoforms, CDI Labs' HuProt Human Proteome Microarray is the most comprehensive human full-length protein array ever created. This covers over 81% of human proteins in each major functional category of the proteome, as defined by the Human Protein Atlas, and allows hundreds of interactions to be profiled at once in a wide range of applications, including mapping antigen-specific immunity (multi-isotype serum profiling), determining monoclonal antibody specificity, and to study protein-protein interaction, substrate identification, protein-DNA binding, protein-RNA binding, and binding of some small molecules.

The HuProt expression library was created by inserting full-length human open reading frames (ORFs) inserted into a yeast high-copy expression vector that produces GST-His6 fusion proteins when induced in yeast. Each human ORF expression vector was verified for integrity by Sanger sequencing. These sequence-verified expression vectors encoding human ORFs were transformed into yeast to create our master library; this eukaryotic expression system ensures HuProt™ proteins maintain both function and proper conformation.

Full-length synthesized proteins are purified via their GST-His6 fusion tags. These proteins are then frozen in aliquots in a 384-well format for later use in printing on HuProt microarrays.

To print individual batches of HuProt arrays, the purified human proteins are then printed as duplicate spot pairs using an Arrayjet UltraMarathon printer (Arrayjet, UK) on PATH nitrocellulose slides (GraceBio, USA). Controls on HuProt include titrated GST protein, titrated human IgG, other human antibody isotypes, histones, mouse and rabbit anti-biotin, mouse IgM, and biotin-tagged control for streptavidin detection. HuProt also contains landmark control spots, including Alex Fluor 555/647 (Jeong *et. al.*, 2012). Success of each batch of HuProt microarrays is validated by anti-GST staining. This confirms full-length GST-His6 fusion proteins were successfully expressed, synthesized, purified, and printed across our clone collection.

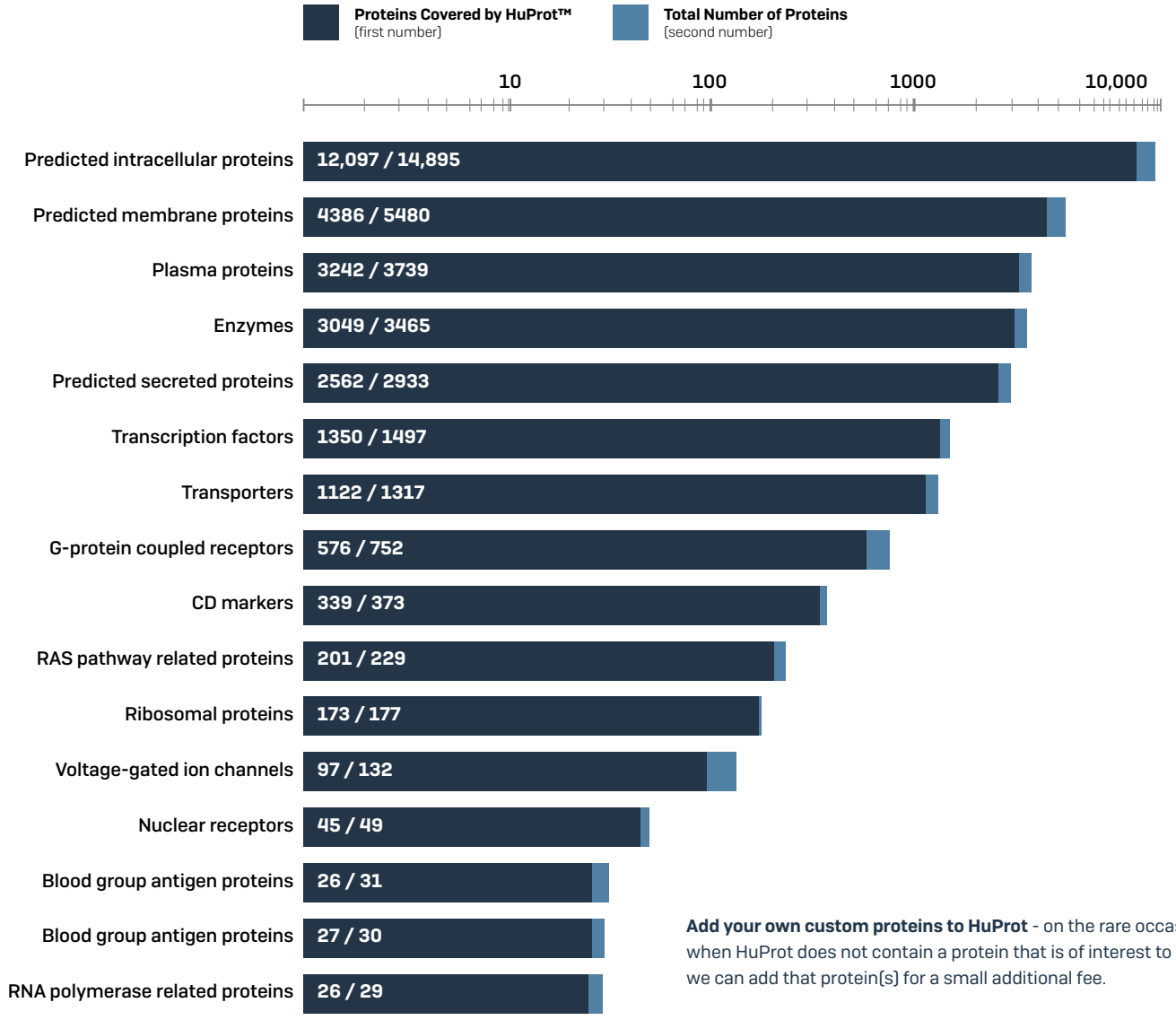
Sample Requirements

Human serum or plasma	**One 20 µL per sample**
Antibody isolates	**10 µg per sample** (concentration 0.1 mg/mL)
Human (CSF) cerebrospinal fluid	**Minimum 1.5 mL per sample**
Protein / peptide / small molecule	**10 µg per sample** (concentration 0.1 mg/mL)
RNA	**1 mL per sample** (concentration 2 µM)
Other fluids	**> Contact us** cdilabs.com/**contact**

HuProt™ Human Proteome Microarray

The Most Comprehensive Collection of Full-length, Correctly-folded 3D Proteins

HuProt consists of more than 21,000 unique human proteins, isoform variants and protein fragments – covering 16,794 unique genes, it is the world's most comprehensive collection of full-length human proteins.

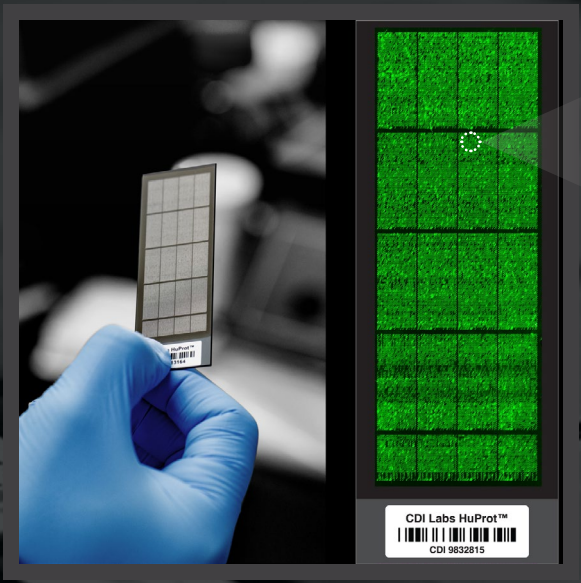


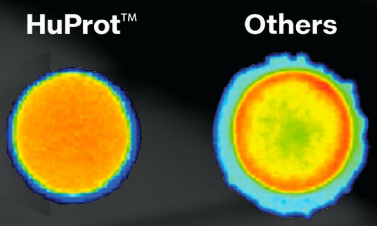
Add your own custom proteins to HuProt - on the rare occasion when HuProt does not contain a protein that is of interest to you, we can add that protein(s) for a small additional fee.

Content includes major functional classes such as intracellular proteins, membrane proteins, enzymes, secreted proteins, transcription factors, transporters, GPCRs, cytokines, immune receptors, immune checkpoints, CD markers, ion channels, cytosolic proteins, nuclear receptors. Additionally there is thorough coverage for proteins enriched in major tissues of interest such as testis, cerebral cortex, thyroid gland, skin, fallopian tube, liver, parathyroid, intestine, kidney, spleen, muscle, epididymis, lymph node, bone marrow, adrenal gland, esophagus, heart, appendix, tonsil, prostate, rectum, adipose tissue, stomach, colon, cervix, uterus, gallbladder, seminal vesicle, breast, ovary, endometrium, smooth muscle, salivary gland, pancreas, and bladder.

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Reproducible Protein Distribution





CDI Labs' non-contact piezoelectric 'inkjet' process uses next-generation Arrayjet printers and allows for rapid production of high quality microarray slides time after time. In comparison to older contact pin printing methods – HuProt arrays are made with improved accuracy and reproducibility with excellent spot morphology.

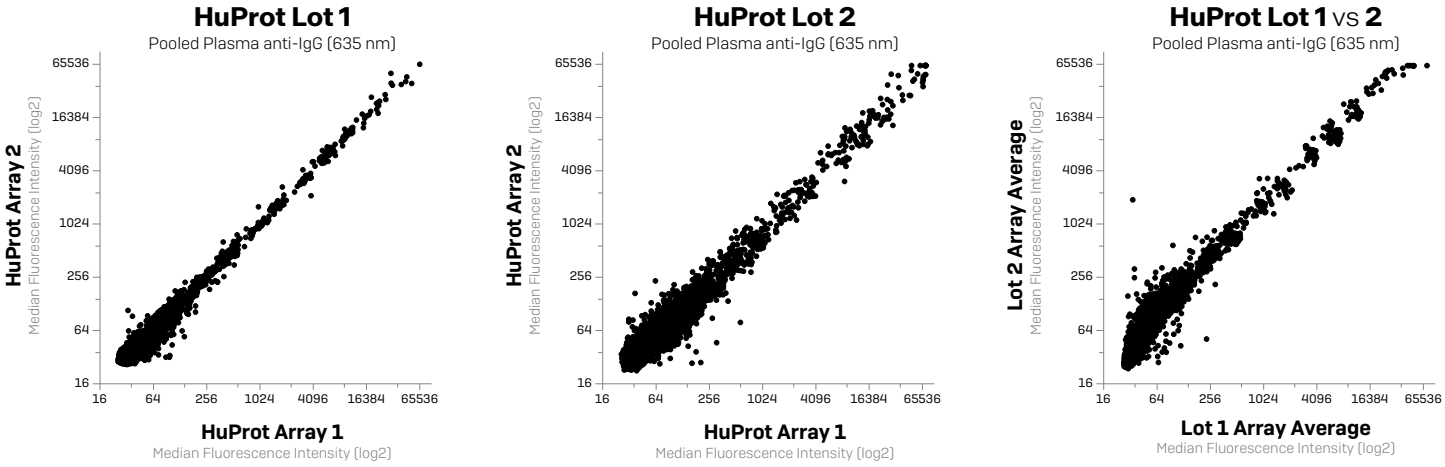
Full-length, Correctly-folded Proteins in Native Conformation

Protein purification prior to microarray manufacturing selects for proteins with correctly folded GST-fusion proteins. Testing on the HuProt Microarray demonstrates that the proteins are folded in native conformation and retain function. When both native and denatured HuProt microarrays were probed with monoclonal antibodies that selectively recognize either linear or folded epitopes of their cognate antigen, the antibodies were found to recognize the appropriate antigen form[2]. Further tests (RNA binding) demonstrated that proteins on HuProt do retain function.

Reproducible Serum and Plasma Profiling

Reproducible Proteome-wide IgG Autoantibody Immunoprofiling of Healthy Human Pooled Plasma

Within and Across HuProt Proteome Microarray Batches



Pooled human plasma was incubated on pairs of HuProt proteome microarrays across two print batches and stained with a fluorescent anti-IgG secondary antibody. Raw median fluorescence intensity data were captured, log2 transformed, and plotted. Linear regression analysis demonstrates reproducibility across the array pair from Lot 1 (R^2 = 0.96), Lot 2 (R^2 = 0.92), and the averaged data from both array batches (R^2 = 0.96). These results demonstrate the robust reproducibility of HuProt autoantibody seromics data both within and across microarray production lots.

HuProt™ Focused Arrays
for Biomarker Validation

CDI Labs offers high-throughput autoantibody signature validation services using focused arrays. Autoantibody biomarkers targeting up to 300 proteins discovered using HuProt can be arrayed in a format optimized for the size of your biomarker set and throughput requirements. This service offers a fast and affordable solution for biomarker validation.

CDI Labs also synthesizes individual proteins at scale for high-throughput screening and other applications.

Contact us at cdilabs.com/**contact** for your research needs.

HuProt™ – a Transformative Product for Autoantibody Profiling

Product Feature	Benefits
21,000+ full-length, corrected-folded 3D human proteins and isoform variants	Access to the most comprehensive full-length human protein collection currently available
High-sensitivity	As little as 3 µLs of serum or plasma required
Protein purification process ensures correctly folded GST-tags	Detection of conformationally specific autoantibodies
Broad coverage across all protein subclasses	Unbiased content for true discovery applications
Single or dual color analysis available	Multi-isotype analysis on a single array
Cited 250+ publications	Performance demonstrated by the research community across many applications
Available for sale or can be processed as a service	Cost effective solutions for both small and large projects
Custom focused arrays available	Higher throughput and more cost effective for more targeted or validation experiments
Individual proteins available	A single protein (platform) can be used for discovery and highly targeted studies

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VirD™ Functional Membrane Protein Array

400+ Unique Proteins for Drug Development

G protein-coupled receptors (GPCRs) play critical roles in a wide variety of physiological processes, and mutations in GPCRs are implicated in many diseases. GPCR receptors must be embedded in a cell membrane to fold correctly and are difficult to express in traditional recombinant expression systems. The most common method used to study GPCRs is to over-express them in transfected mammalian cell lines. However, the transfected cells also contain endogenous receptors, making it a challenge to interpret experimental results.

Our VirD viral-based expression technology expresses individual GPCR proteins embedded in a cell membrane. Purified virions displaying correctly-folded GPCRs are printed onto glass to create VirD arrays. This method eliminates the problem of endogenous receptor competition when using traditional methods, and yields much cleaner data. VirD GPCR arrays can be purchased directly or provided as a complete analysis service.

VirD Membrane Protein Array Validation and Production

VirD is a microarray that uses virions to display functional membrane proteins like GPCRs and is useful for drug development and functional assays.



1 Human G protein-coupled receptor (GPCR) open reading frames (ORFs) modified and cloned

2 Engineered virion clones are used to infect mammalian cells

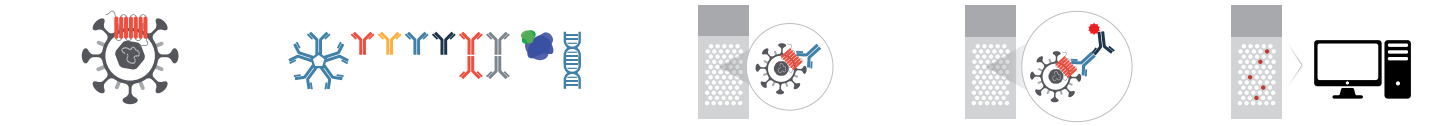
3 Transfected virions bearing V5 tags are bi-directionally sequenced

4 Arrays are printed in batches of up to 600

5 Two slides from each batch are stained for HSV-1 protein gD and scanned for QC

VirD Membrane Protein Array Methods and Analysis

VirD is a sensitive membrane protein matrix platform that provides quantitative binding interaction data from a few microliters of sample. VirD can help identify the function of orphan GPCRs and can be used to develop new drugs targeting membrane proteins and receptors.



1 Each sequence-confirmed HSV-1 virion is checked for V5 tag

2 VirD can detect most any target of interest

3 Samples of interest are diluted and reacted with array

4 Samples of interest are labeled and scanned

5 VirD library file is aligned and spot QC-ed

Technical Details

To create the VirD array, 500+ unique human membrane protein open reading frames (ORFs) are modified to remove their stop codons and subcloned into the UL27 locus of the herpes simplex (HSV-1) virus genome with a C-terminal v5 tag. These individually engineered viron clones are then used to infect mammalian cells. Native human protein assembly machinery correctly folds the proteins and shuttles them to the viral membrane. Membrane-protein bearing VirD virions are concentrated from culture supernatants and their membrane-protein expression confirmed via anti-V5 immunoblots.

Purified virions were then printed on SuperEpoxy glass slides to create the final GPCR-VirD arrays. The quality of printed VirD-GPCR arrays is confirmed via staining for HSV-1 Glycoprotein D (gD) via anti-gD antibody and comparing those as compared with bovine serum albumin negative controls (BSA).

Sample Requirements

Antibody isolates	**10 µg per sample** (concentration 0.1 mg/mL)
Protein / peptide / small molecule	**10 µg per sample** (concentration 0.1 mg/mL)
RNA	**1 mL per sample** (concentration 2 µM)
Other fluids	**> Contact us** cdilabs.com/**contact**

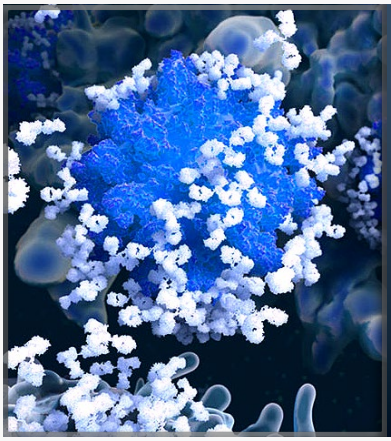
Human serum or plasma often contains anti-HSV-1 antibodies making VirD inappropriate for clinical studies.

Disease Areas

Transformational Tools and Services for Disease Research and Drug Development

CDI Labs provides transformational tools and services for proteome-wide autoantibody seromics and target identification applications. Products such as the HuProt™ Human Proteome Array, the world's largest collection of full-length proteins, and VirScan™*, HuScan®* and MouseScan™* , a portfolio of services that use bacteriophage immunoprecipitation sequencing (PhIP-Seq), have been used in a wide range of applications by the research and drug development community. These applications have encompassed a broad range of disease areas including autoimmune diseases, oncology, neurological diseases, infectious diseases, genetic diseases and the study of aging.

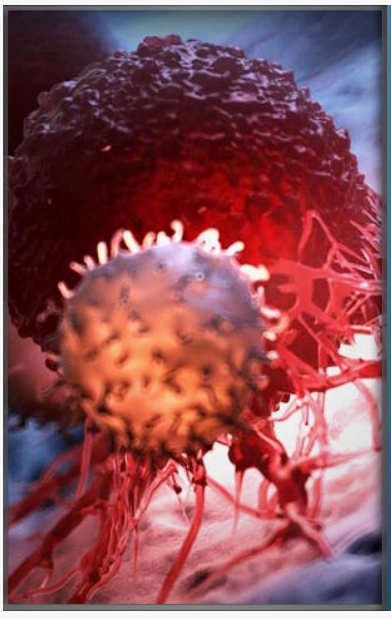
* VirScan, HuScan and MouseScan are provided exclusively by CDI Labs Canada.



Autoimmune Disease

Autoantibodies play a crucial role in the development and progression of autoimmune diseases which occur when the immune system mistakenly targets and attacks the body's own tissues, mistaking them for foreign invaders.

Understanding the specific autoantibodies involved in an autoimmune disease can guide treatment strategies. For example, therapies may aim to suppress the overall immune response, reduce inflammation, or target specific components of the immune system responsible for producing autoantibodies.



Oncology

Cancer research and treatment has been revolutionized with the advent of immune-based cancer therapy, also known as immunotherapy or immuno-oncology (IO). Unlike traditional cancer treatments which aim to directly target cancer cells, immunotherapy works by enhancing or restoring the immune system's ability to detect and attack cancer cells. Checkpoint inhibitors are an example of this approach, and autoantibodies signatures have demonstrated the potential to predict both disease recurrence and immune-related adverse events in melanoma patients treated with checkpoint inhibitor adjuvant immunotherapy.

In addition to serum profiling applications, HuProt is an invaluable target identification tool which enables discovery of diverse tumor-associated targets at a proteome-wide scale. Applications include:

- **Development and specificity testing of antibody-drug conjugates** (ADCs)
- **Development of multi-specific medicines which overcome the challenge of "undruggable targets" by using the principle of chemically induced proximity** (CIP)
- **Identification of small molecule drug targets which can be discovered on HuProt via biotinylation of the small molecule being studied.**

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Disease Areas



Neurological Disease

Autoantibodies can play a significant role in neurological diseases, contributing to various autoimmune disorders that affect the central nervous system. In these conditions, the immune system mistakenly targets components of the nervous system, leading to inflammation, tissue damage, and neurological symptoms.

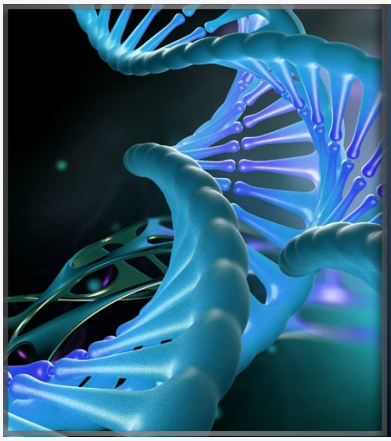
In addition to autoantibody seromics, CDI Labs products and services can also be used to identify novel targets and pathways (target identification) via applications such as protein-protein interaction studies. These approaches can lead to a better understanding of neurological diseases and potentially inform new therapeutics and treatments options.



Infectious Disease

Autoantibodies can sometimes be associated with infectious diseases, and their presence or induction is a complex aspect of the host immune response. Infections have the capability of inducing a new autoimmune response or worsening an existing condition.

Molecular mimicry is a phenomenon where the antigens of infectious agents resemble host tissues. This can lead to the production of antibodies that cross-react with both the infectious agent and self-antigens. For example, research indicates that Epstein-Barr virus can lead to the development of MS via high-affinity molecular mimicry between the EBV transcription factor EBV nuclear antigen 1 (EBNA1) and the central nervous system protein glial cell adhesion molecule (GlialCAM).



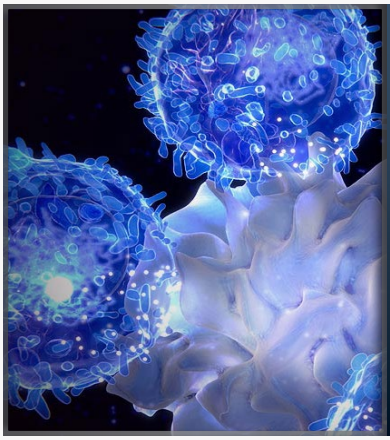
Genetic Disease

While many genetic conditions are not explicitly thought of as immunologic disorders, it is impossible to separate the immune system from the rest of the human body. The adaptive immune system in particular, representing approximating half of all nucleated cells in the human body, has behavior that is particularly sensitive to genetic dysregulation. These effects can create over-active immunity with increased rates of autoimmune and inflammatory disease, or under-active immunity that makes the affected patients more susceptible to infection.

Applications | **Autoantibody Seromics**

What is Autoantibody Seromics?

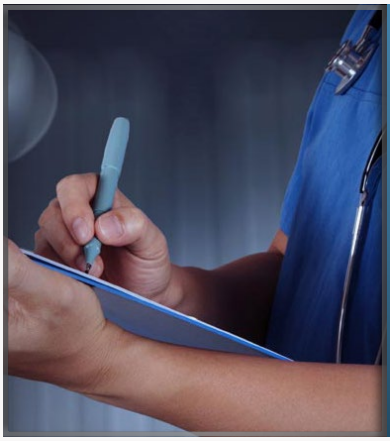
Autoantibody seromics refers to the systematic study of autoantibodies present in the blood serum of individuals. Autoantibodies are antibodies that target the body's own tissues and cells, potentially leading to autoimmune diseases. By analyzing autoantibody profiles, researchers aim to uncover biomarkers that can aid in the early detection and monitoring of autoimmune and other diseases, as well as identify potential targets for personalized therapies aimed at modulating the immune response. Additionally, autoantibody seromics can contribute to our understanding of the underlying immunological processes involved in disease, paving the way for the development of novel treatment strategies.



Biomarker Discovery

Autoantibodies serve as valuable biomarkers for a wide range of diseases and conditions. As biomarkers, autoantibodies offer several advantages, including their specificity, accessibility, and potential for early detection. The identification of autoantibody biomarkers or signatures can provide researchers with highly valuable information relevant to many aspects of human disease, including:

- Disease mechanisms
- Disease classification
- Prediction of therapy response
- Likelihood of adverse treatment reactions
- Diagnostic information



Patient Stratification

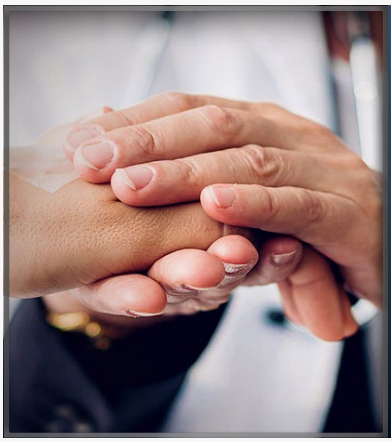
Patient stratification involves categorizing individuals with a particular disease or condition into subgroups based on the presence or absence of specific autoantibodies. For example, patients with certain autoantibodies may have a higher risk of developing complications or experiencing disease flares, as is the case with systemic lupus erythematosus (SLE). This stratification approach can provide information that can help:

- Tailor treatment strategies
- Predict disease progression
- Identify individuals at higher risk of complications
- Identify individuals at higher risk of disease recurrence

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Applications | **Autoantibody Seromics**



Patient Response and Adverse Reaction Prediction

Autoantibodies can play a role in predicting patient response to treatment. Combining multiple autoantibodies into biomarker panels or signatures can improve the accuracy of stratification and predict both treatment response and immune-related adverse events (irAEs). Integrating autoantibody data with other clinical and molecular markers can enhance risk prediction models and facilitate personalized treatment approaches.



Diagnostic Development

The presence of specific autoantibodies can be a key diagnostic criterion for certain diseases. Autoantibody panels consisting of specific autoantibodies have the potential to be used for the diagnosis and classification of autoimmune diseases where diagnosis is challenging due to the heterogeneity of the clinical course, symptoms and severity of the disease. Autoantibodies are frequently detected in the serum or plasma of patients with a wide range of diseases, including:

- **Autoimmune diseases such as rheumatoid arthritis**
- **Systemic lupus erythematosus (SLE)**
- **Autoimmune thyroid diseases** *(e.g., Hashimoto's thyroiditis, Graves' disease)*
- **Type 1 diabetes**



Vaccine Development and Safety

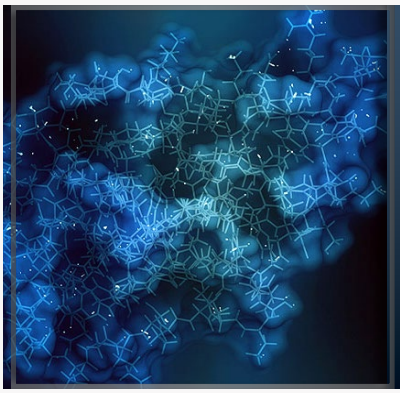
Measurement of vaccine-induced autoantibody responses can provide insights into the magnitude and specificity of the immune response elicited by a vaccine. Autoantibodies may also be involved in the safety evaluation of vaccines by assessing the risk of autoimmune reactions or immune-related adverse events (irAEs) associated with vaccination.

Cancer vaccines work by stimulating the body's immune system to recognize and attack cancer cells specifically. These vaccines are designed to stimulate the immune system to trigger an anti-tumor response by utilizing tumor antigens.

Applications | **Target Identification**

HuProt™ Can Identify the Biological Pathways and Mechanisms Underlying Human Disease

Target identification (ID) is a crucial step in the drug discovery and development process. It is aimed at identifying specific molecules or biological pathways that play a key role in a disease process and can be targeted by therapeutic interventions. HuProt™ is a powerful tool for target identification as it allows researchers to identify how a compound or molecule of interest interacts with human proteins at a proteome-wide scale, in a single experiment. HuProt can be used to asses protein binding to a wide variety of compounds or molecules including, proteins (protein-protein interactions), DNA, RNA, antibodies (antibody specificity studies), and many more.



Protein-Protein Interactions

Protein-protein interactions (PPIs) are essential for virtually all biological processes. PPIs occur when two or more proteins bind together to form complexes, enabling various cellular activities. Understanding and targeting PPIs is a key strategy in today's drug discovery process. Dysregulated PPIs are implicated in numerous diseases, including:

- Cancer
- Neurodegenerative disorders
- Infectious diseases
- Autoimmune diseases



Protein-Nucleic Acid Interactions

Protein-nucleic acid interactions involve proteins binding to nucleic acids, such as DNA or RNA, to regulate their function and activity. Understanding protein-nucleic acid interactions and developing strategies to modulate these interactions hold great promise for advancing the understanding of gene regulation and for developing novel therapeutics. Protein-nucleic acid interactions play a fundamental role in various biological processes, including:

- Gene expression
- DNA replication
- Transcription
- Translation

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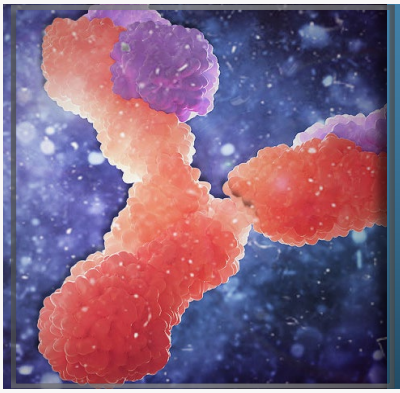
Applications | **Target Identification**



Target Deconvolution / **Target Discovery**

Target deconvolution is a process in drug discovery and development that involves identifying the specific molecular target(s) of a monoclonal or polyclonal antibody, bioactive compound or drug candidate. For example, monoclonal antibodies with unknown targets may have therapeutic potential for treating diseases or conditions for which the target antigen is unknown or poorly characterized. Target deconvolution is crucial for understanding:

- **The mechanism of action of an antibody, compound or drug candidate**
- **Elucidating a drug candidate's effect on biological pathways**
- **Assessing potential therapeutic applications**



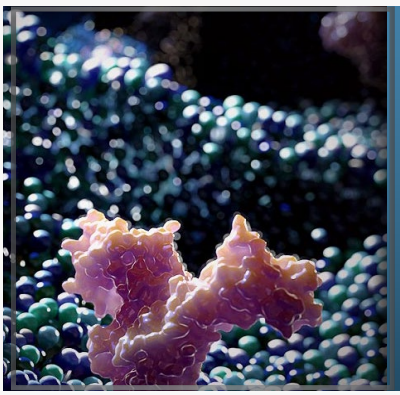
Antibody Specificity Validation

Monoclonal antibody specificity validation is a crucial step in antibody development. Specificity testing validates that the antibody recognizes its intended target with high specificity while exhibiting minimal cross-reactivity with unrelated proteins or molecules.



Post-translational Modification Specificity Testing

Post-translational modifications (PTMs) are chemical modifications that occur on proteins after they have been synthesized. PTMs regulate protein activity, localization, and interactions by altering protein structure, stability, and interactions with other molecules. As PTMs play crucial roles in protein function, targeting PTMs has emerged as a promising strategy in drug discovery. PTM specificity testing is crucial for validating binding to modified forms of proteins.



Chemically Induced Proximity (CIP)

Many disease treatments target a specific protein with the goal of blocking, activating, or destroying it (often called targeted degradation). The challenge with this methodology is that most human proteins are "undruggable" with conventional medicines. Multispecific medicines overcome this challenge by using the principle of chemically induced proximity (CIP), in which they bring two things together by using small molecules or chemical ligands. CIP technologies that leverage small molecules (or chemical ligands) to induce the proximity of target proteins with specific cellular components include:

- **PROTACs** (PROteolysis TArgeting Chimeras)
- **AUTOTACs** (AUTOphagy- TArgeting Chimeras)
- **LYTACs** (LYsosome TArgeting Chimeras)

Custom Monoclonal Antibody Development and Validation

20,000+ Antigens Available

Using HuProt™ Human Proteome Microarray to validate the final product, your antibody is screened for cross-reactivity against the largest collection of human proteins on a single slide – representing approximately 75% of the human proteome.

CDI Labs' monoclonal antibody development, production and purification, along with HuProt microarray specificity validation, is (approximately) an 8-week process. It is a milestone-driven, 4-phase process with client reporting and assessment exchanged at each step. Invoicing begins at the initiation of Phase 1 and at the completion of each phase thereafter. Milestones represent potential endpoints that depend on your project requirements or if the results-to-date fail to meet your expectations.

Monospecificity in 8 Weeks

Our antibody development platform addresses the urgent need for monoclonal antibodies with unprecedented specificity to native or denatured proteins.

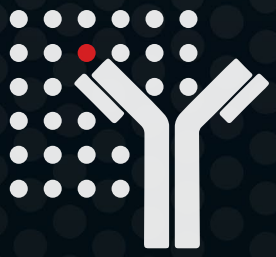
Quote	Day 1	Day 14	Day 38	Day 52	Day 56
Project Specification Antigen design, immunization strategy and screening methodology are critical to achieving high-value results. CDI Labs offers a fully-customized development process to meet the requirements and provide a quote based on deliverables.	**Phase 1** **Antigen and Immunization** 5-mouse immunization regimen using as little as 200 µg client-supplied protein of >80% purity or carrier-conjugated peptide. We also offer assistance in antigen design and synthesis.	**Phase 2** **Fusion and Screening** 2 top-responding animals are chosen, and immune cells are harvested for fusion with myeloma cells yielding multiple hybridomas. Supernatants from IgG-secreting clones are screened by ELISA.	**Phase 3** **Sub-cloning, Expansion and Specificity Testing** A minimum of 5 to a maximum of 30 high-quality clones are chosen and sub-cloned. Each is expanded into 24-well plates and subsequently into T-25 flasks. Supernatants are shipped for further client evaluation.	**Phase 4** **Production and Purification** Each clone is vialed, labeled and frozen. Samples of each are reserved for mycoplasma testing. Per client instructions, 3 select clones are placed into antibody production. The resulting antibodies are purified and evaluated with HuProt Human Proteome Microarray.	**Deliverables** Detailed specificity data and additional QA documentation are prepared and shipped along with the purified antibodies and all frozen clones.

Antibodies Come in a Variety of Forms Called Isotypes

Depending on the isotype, antibodies can promote cell killing, protective immune tolerance, or help educate T cells.

	IgM		IgG				IgA				IgD		IgE
Description	Prior to class switching or affinity maturation, the primary antibody response is composed of IgM. Common, with strong complement fixation.		Most abundant serum antibody. Can induce cytotoxic phagocytic responses and encourage cytotoxic T cell activation via targeting to antigen presenting cells. Can be tolerogenic depending on subclass when bound to FcγRIIB. Crosses placenta.				Dimeric form of IgA2 is the most abundant secreted antibody across mucous membranes such as in the mouth and colon as well as breast milk. Monomers of IgA1 are more common in serum.				Membrane-bound IgD is a hallmark of mature B cells. Found in serum in low amounts, no consensus on exact serum function		Rare antibody with extreme ability to activate FcεR allergic responses. Evolved for parasite defense.
	 IgM B cell Receptor	 IgM Pentamer	 IgG1	 IgG2	 IgG3	 IgG4	 Secreted IgA1 Dimer	 IgA1	 Secreted IgA2 Dimer	 IgA2	 IgD B cell Receptor	 IgD	 IgE
Approximate % of total serum antibody	0	10%	53%	15%	6%	4%	0	11%	0	1%	0	0.2%	0.004%
Approximate serum half-life (days)	0	5–10	21	21	7	21	0	5–7	0	4–6	0	2–8	1–5
Often cytotoxic activation by immune complex FcγRIIA	–	–					–	–	–	–	–	–	–
Often tolerogenic inhibition by immune complex FcγRIIB	–	–		–			–	–	–	–	–	–	–
Activation of classical complement pathway (C1q)	–					–	–		–	–	–	–	–
Ability to cross the placenta during pregnancy	–	–					–	–	–	–	–	–	–
Dominant presence in breast milk and mucous membranes	–	–	–	–	–	–		–		–	–	–	–
Single molecule FcεR activation; response to parasites and allergens	–	–	–	–	–	–	–	–	–	–	–	–	

Chart prepared by Tyler Hulett, PhD *with references (1-3). References* ***1.*** *Nimmerjahn, F. Divergent Immunoglobulin G Subclass Activity Through Selective Fc Receptor Binding. Science 310, 1510-1512 (2005).*
2. *Vidarsson, G. Dekkers, G & Rispens, T. IgG Subclasses and Allotypes From Structure to Effector Functions. Front. Immuno. 5, (2014).* ***3.*** *Fundamental Immunology. (Walters Kluwer/Uppincott Williams & Wilkins, 2008).*



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